Ordinary Differential Equations

Chris Ritter

September 15, 2026

Contents

1 First-Order Equations **2**

1.1 Exact Differential Equations . . . 2

1.2 Example . . . 3

1.3 Checking Whether an Equation Is Exact . . . 4

1.4 Finding the Potential Function . . . 5

1.5 Example: Checking Exactness and Solving . . . 6

2 Integrating Factors **7**

2.1 Integrating Factor Depending Only on x . . . 8

2.2 Integrating Factor Depending Only on y . . . 8

2.3 Example: Finding an Integrating Factor . . . 9

2.3.1 Making the Equation Exact . . . 11

2.3.2 Finding the Solution . . . 11

2.4 Integrating Factor Summary . . . 12

3 Summary **13**

3.1 Exact Equations: Standard Procedure . . . 13

3.2 Integrating Factors: Standard Procedure . . . 14

3.3 Big Picture . . . 16

These notes introduce several important classes of first-order differential equations, with particular emphasis on exact equations and integrating factors.

1 First-Order Equations

Some important types of first-order differential equations are:

- Linear equations
- Separable equations
- Exact equations

1.1 Exact Differential Equations

A first-order differential equation can be written in the form

$$M(x,y) + N(x,y)\frac{dy}{dx} = 0.$$

Definition: Exact Differential Equation.

The differential equation

$$M(x,y) + N(x,y)\frac{dy}{dx} = 0$$

is called **exact** if there exists a function $\psi = \psi(x,y)$ such that

$$\psi_x = M, \qquad \psi_y = N.$$

The function $\psi(x,y)$ is called a **potential function**.

If such a function ψ exists, then by the chain rule,

$$\frac{d}{dx}\psi(x,y(x)) = \psi_x + \psi_y\frac{dy}{dx}.$$

Since

$$\psi_x = M$$

and

$$\psi_y = N,$$

we obtain

$$\frac{d}{dx}\psi(x,y) = M(x,y) + N(x,y)\frac{dy}{dx}.$$

Therefore, if

$$M(x,y) + N(x,y)\frac{dy}{dx} = 0,$$

then

$$\frac{d}{dx}\psi(x,y) = 0.$$

It follows that

$$\boxed{\psi(x,y) = C}.$$

Key idea.

For an exact differential equation, the main goal is to find the potential function $\psi(x,y)$. Once ψ is found, the solution is

$$\boxed{\psi(x,y) = C}.$$

1.2 Example

Example: Finding the Potential Function.

Consider

$$2x + y^2 + 2xy\frac{dy}{dx} = 0.$$

Identify

$$M(x,y) = 2x + y^2, \qquad N(x,y) = 2xy.$$

We seek $\psi(x,y)$ satisfying

$$\psi_x = 2x + y^2, \qquad \psi_y = 2xy.$$

Integrating ψ_x with respect to x gives

$$\psi(x,y) = \int (2x + y^2)\,dx + h(y),$$

so

$$\psi(x,y) = x^2 + xy^2 + h(y).$$

Differentiate with respect to y:

$$\psi_y = 2xy + h'(y).$$

Since $\psi_y = N = 2xy$,

$$2xy + h'(y) = 2xy,$$

and therefore

$$h'(y) = 0.$$

Thus,

$$h(y) = C.$$

Absorbing the constant into the right-hand side gives

$$\boxed{x^2 + xy^2 = C}.$$

1.3 Checking Whether an Equation Is Exact

The following theorem gives a convenient test for exactness.

Theorem: Exactness Criterion.

Let M, N, M_y, and N_x be continuous in a rectangular region R. Then

$$M(x,y) + N(x,y)\frac{dy}{dx} = 0$$

is exact if and only if

$$\boxed{M_y = N_x}$$

for all $(x,y) \in R$.

To see why this condition appears, suppose that the equation is exact. Then there exists $\psi(x, y)$ such that

$$\psi_x = M, \qquad \psi_y = N.$$

Taking mixed partial derivatives,

$$\psi_{xy} = M_y$$

and

$$\psi_{yx} = N_x.$$

Because the relevant partial derivatives are continuous,

$$\psi_{xy} = \psi_{yx}.$$

Hence,

$$\boxed{M_y = N_x}.$$

1.4 Finding the Potential Function

Suppose

$$M(x, y) + N(x, y)\frac{dy}{dx} = 0$$

is exact.

Since

$$\psi_x = M,$$

we integrate M with respect to x:

$$\boxed{\psi(x, y) = \int M(x, y)\, dx + h(y)}.$$

The function $h(y)$ appears because we are integrating with respect to x. Any term depending only on y has derivative zero with respect to x.

We then differentiate with respect to y:

$$\psi_y = \frac{\partial}{\partial y}\left(\int M(x, y)\, dx\right) + h'(y).$$

Since $\psi_y = N$,

$$h'(y) = N(x,y) - \frac{\partial}{\partial y}\left(\int M(x,y)\,dx\right).$$

Finally, integrate $h'(y)$ to obtain $h(y)$ and therefore $\psi(x,y)$.

1.5 Example: Checking Exactness and Solving

Example: Exact Equation.

Consider

$$(y\cos x + 2xe^y) + \left(\sin x + x^2e^y - 1\right)\frac{dy}{dx} = 0.$$

Identify

$$M(x,y) = y\cos x + 2xe^y$$

and

$$N(x,y) = \sin x + x^2e^y - 1.$$

First,

$$M_y = \cos x + 2xe^y.$$

Also,

$$N_x = \cos x + 2xe^y.$$

Therefore,

$$M_y = N_x,$$

so the equation is exact.

Since

$$\psi_x = y\cos x + 2xe^y,$$

integrate with respect to x:

$$\psi(x,y) = y\sin x + x^2e^y + h(y).$$

Differentiate with respect to y:

$$\psi_y = \sin x + x^2 e^y + h'(y).$$

Set this equal to N:

$$\sin x + x^2 e^y + h'(y) = \sin x + x^2 e^y - 1.$$

Thus,

$$h'(y) = -1, \qquad h(y) = -y + C.$$

Hence,

$$\boxed{y \sin x + x^2 e^y - y = C}.$$

2 Integrating Factors

An equation of the form

$$M(x, y) + N(x, y)\frac{dy}{dx} = 0$$

may not be exact. We can sometimes multiply the equation by a function $\mu(x, y)$ so that the resulting equation becomes exact.

Definition: Integrating Factor.

An **integrating factor** is a function $\mu(x, y)$ that can be multiplied into a differential equation so that the resulting equation is exact.

We want

$$\mu M + \mu N \frac{dy}{dx} = 0$$

to be exact. Therefore,

$$\frac{\partial}{\partial y}(\mu M) = \frac{\partial}{\partial x}(\mu N).$$

Using the product rule,

$$\mu_y M + \mu M_y = \mu_x N + \mu N_x.$$

2.1 Integrating Factor Depending Only on x

Suppose

$$\mu = \mu(x).$$

Then

$$\mu_y = 0.$$

Therefore,

$$\mu M_y = \mu_x N + \mu N_x.$$

Rearranging,

$$\mu(M_y - N_x) = \mu_x N.$$

Thus,

$$\frac{\mu_x}{\mu} = \frac{M_y - N_x}{N}.$$

Therefore,

$$\boxed{\frac{d\mu}{\mu} = \frac{M_y - N_x}{N}\, dx}.$$

If the right-hand side is a function of x alone, then

$$\boxed{\mu(x) = e^{\int \frac{M_y - N_x}{N}\, dx}}.$$

2.2 Integrating Factor Depending Only on y

Similarly, suppose

$$\mu = \mu(y).$$

Then

$$\mu_x = 0.$$

Therefore,

$$\mu_y M + \mu M_y = \mu N_x.$$

Rearranging,

$$\mu_y M = \mu(N_x - M_y).$$

Thus,

$$\frac{\mu_y}{\mu} = -\frac{M_y - N_x}{M}.$$

Therefore,

$$\boxed{\frac{d\mu}{\mu} = -\frac{M_y - N_x}{M}\, dy}.$$

If the right-hand side is a function of y alone, then

$$\boxed{\mu(y) = e^{-\int \frac{M_y - N_x}{M}\, dy}}.$$

2.3 Example: Finding an Integrating Factor

Example: Non-Exact Equation.

Consider

$$(3xy + y^2) + (x^2 + xy)\frac{dy}{dx} = 0.$$

Identify

$$M(x, y) = 3xy + y^2, \qquad N(x, y) = x^2 + xy.$$

First,

$$M_y = 3x + 2y$$

and

$$N_x = 2x + y.$$

Since

$$M_y \neq N_x,$$

the equation is **not exact**.

We look for an integrating factor depending only on x.

Thus,

$$\frac{d\mu}{\mu} = \frac{M_y - N_x}{N}\,dx.$$

Substitute:

$$\frac{d\mu}{\mu} = \frac{(3x+2y)-(2x+y)}{x^2+xy}\,dx.$$

Simplifying,

$$\frac{d\mu}{\mu} = \frac{x+y}{x^2+xy}\,dx.$$

Factor the denominator:

$$x^2 + xy = x(x+y).$$

Therefore,

$$\frac{d\mu}{\mu} = \frac{x+y}{x(x+y)}\,dx = \frac{1}{x}\,dx.$$

Integrating,

$$\int \frac{d\mu}{\mu} = \int \frac{1}{x}\,dx.$$

Hence,

$$\ln|\mu| = \ln|x| + C.$$

Up to a nonzero constant multiple,

$$\boxed{\mu(x) = x}.$$

2.3.1 Making the Equation Exact

Multiply the original equation by $\mu = x$:

$$x(3xy + y^2) + x(x^2 + xy)\frac{dy}{dx} = 0.$$

Thus,

$$(3x^2y + xy^2) + (x^3 + x^2y)\frac{dy}{dx} = 0.$$

Define

$$\widetilde{M}(x, y) = 3x^2y + xy^2$$

and

$$\widetilde{N}(x, y) = x^3 + x^2y.$$

Now,

$$\widetilde{M}_y = 3x^2 + 2xy$$

and

$$\widetilde{N}_x = 3x^2 + 2xy.$$

Therefore,

$$\widetilde{M}_y = \widetilde{N}_x,$$

so the new equation is exact.

2.3.2 Finding the Solution

Since

$$\psi_x = 3x^2y + xy^2,$$

integrate with respect to x:

$$\psi(x, y) = \int (3x^2y + xy^2)\, dx + h(y).$$

Thus,

$$\psi(x,y) = x^3 y + \frac{x^2 y^2}{2} + h(y).$$

Differentiate with respect to y:

$$\psi_y = x^3 + x^2 y + h'(y).$$

Since

$$\psi_y = \widetilde{N} = x^3 + x^2 y,$$

we obtain

$$h'(y) = 0.$$

Therefore,

$$h(y) = C.$$

Hence,

$$\boxed{x^3 y + \frac{x^2 y^2}{2} = C}.$$

2.4 Integrating Factor Summary

Important formula.

For

$$M(x,y) + N(x,y)\frac{dy}{dx} = 0,$$

if the equation is not exact,

$$M_y \neq N_x,$$

check

$$\boxed{\frac{M_y - N_x}{N}}$$

for dependence on x alone.

If so,

$$\boxed{\mu(x) = e^{\int \frac{M_y - N_x}{N}\,dx}}.$$

Otherwise, check

$$\boxed{-\frac{M_y - N_x}{M}}$$

for dependence on y alone.

If so,

$$\boxed{\mu(y) = e^{-\int \frac{M_y - N_x}{M}\,dy}}.$$

3 Summary

Key idea.

For a differential equation

$$M(x,y) + N(x,y)\frac{dy}{dx} = 0,$$

the first question is whether it is exact.

$$\boxed{M_y = N_x}$$

If yes, find the potential function ψ and solve

$$\boxed{\psi(x,y) = C}.$$

If not, look for an integrating factor μ.

3.1 Exact Equations: Standard Procedure

01. IDENTIFY

Identify $M(x,y)$ and $N(x,y)$:

$$M(x,y) + N(x,y)\frac{dy}{dx} = 0.$$

02. CHECK EXACTNESS

Compute

$$M_y \qquad \text{and} \qquad N_x.$$

Check whether

$$\boxed{M_y = N_x}.$$

03. FIND THE POTENTIAL FUNCTION

If the equation is exact, use

$$\psi_x = M.$$

Integrate with respect to x:

$$\psi(x, y) = \int M(x, y)\, dx + h(y).$$

04. DETERMINE $h(y)$

Differentiate with respect to y and use

$$\psi_y = N.$$

Solve for $h'(y)$ and integrate.

05. WRITE THE SOLUTION

Set the potential function equal to a constant:

$$\boxed{\psi(x, y) = C}.$$

3.2 Integrating Factors: Standard Procedure

If

$$M_y \neq N_x,$$

the equation is not exact.

01. CHECK FOR $\mu(x)$

Compute

$$\frac{M_y - N_x}{N}.$$

If this depends only on x, use

$$\boxed{\mu(x) = e^{\int \frac{M_y - N_x}{N}\,dx}}.$$

02. CHECK FOR $\mu(y)$

If the previous expression does not depend only on x, check

$$-\frac{M_y - N_x}{M}.$$

If this depends only on y, use

$$\boxed{\mu(y) = e^{-\int \frac{M_y - N_x}{M}\,dy}}.$$

03. MULTIPLY BY THE INTEGRATING FACTOR

Multiply the entire differential equation by μ.

Re-identify

$$\widetilde{M} \qquad \text{and} \qquad \widetilde{N}.$$

04. VERIFY EXACTNESS

Check

$$\widetilde{M}_y = \widetilde{N}_x.$$

05. FIND THE POTENTIAL FUNCTION

Find $\psi(x, y)$ using

$$\psi_x = \widetilde{M}.$$

Then use

$$\psi_y = \widetilde{N}$$

to determine the remaining function.

06. WRITE THE SOLUTION

Finally,

$$\boxed{\psi(x, y) = C}.$$

3.3 Big Picture

Key idea.

$$
\begin{array}{c}
M + N\dfrac{dy}{dx} = 0 \\
\downarrow \\
\text{Check } M_y = N_x \\
\downarrow \\
\begin{cases}
\text{Exact} & \longrightarrow \psi \longrightarrow \psi = C \\
\text{Not exact} & \longrightarrow \mu \longrightarrow \text{Exact} \longrightarrow \psi \longrightarrow \psi = C
\end{cases}
\end{array}
$$